

2024, Mar. 2024 Release LOC:Acute Adult**Deep Vein Thrombosis**

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview**Select Day**

Initial review ^(1, 2, 3, 4)

Episode Day 1 ^(1, 3, 4)

Episode Day 2 ^(3, 4, 30)

Episode Day 3 ^(4, 42, 43)

Episode Day 4

Discharge Screens ⁽⁴⁹⁾

Utilization Benchmarks

Notes

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Overview

Instruction: This subset is for patients who have a deep vein thrombosis (DVT) that has been confirmed by imaging. It is intended for patients with an obstructive DVT and excludes superficial vein thrombosis (SVT) unless located in a superficial femoral vein. DVTs can develop in the following veins:

Lower extremity:

- Popliteal
- Peroneal
- Common femoral
- Deep and superficial femoral
- Femoral
- Anterior tibial
- Posterior tibial

Upper extremity:

- Subclavian
- Axillary
- Brachial

Upper extremity DVT can extend to:

- Brachiocephalic vein
- Superior vena cava
- Internal jugular vein

Splanchnic:

- Portal
- Mesenteric
- Splenic

Introduction:

Deep vein thrombosis (DVT) is a condition in which a clot (thrombus) forms within a deep vein and is associated with significant morbidity and mortality if untreated (Ortel et al., Blood Adv 2020, 4: 4693-738). Although DVTs most commonly develop in the lower extremities, they can also occur in the upper extremities or visceral veins.

Evaluation and Treatment:

Clinical symptoms of an extremity DVT may include pain, swelling, erythema, and/or warmth. Symptoms tend to be more severe as the thrombosis extends more proximally causing greater outflow obstruction. Although the symptoms can assist in determining the probability of a DVT, once clinically suspected, imaging should be obtained to confirm the diagnosis (Kakkos et al., *Eur J Vasc Endovasc Surg* 2021, 61: 9-82). Doppler ultrasonography (DUS) is the recommended first-line imaging modality for the confirmation of an extremity DVT (American College of Radiology, ACR Appropriateness Criteria: Suspected Lower Extremity Deep Vein Thrombosis. 2020; American College of Radiology, ACR Appropriateness Criteria: Suspected Upper Extremity Deep Vein Thrombosis. 2020). Splanchnic vein thrombosis, which involves the portal, mesenteric, and/or splenic veins, often has nonspecific clinical manifestations; therefore, the diagnosis relies on imaging. DUS is also the first-line diagnostic test for portal vein thrombosis (PVT). In addition, DUS can identify a splenic vein thrombosis, although it is less sensitive in this condition due to the collaterals that are present near the splenic hilum. Evaluation for a mesenteric vein thrombosis is recommended with either a contrast-enhanced computed tomography (CT) scan or magnetic resonance imaging (MRI) since these modalities better visualize the mesenteric veins (Simonetto et al., *Am J Gastroenterol* 2020, 115: 18-40; Valeriani et al., *Vasc Health Risk Manag* 2019, 15: 449-61).

Anticoagulation therapy is recommended for the treatment of extremity DVTs to prevent progression, dissolution, and recurrence of the thrombus. Direct oral anticoagulants (DOACs) are suggested over vitamin K antagonists (VKA) due to their efficacy, reduced need for monitoring, lack of parenteral overlap with heparin for some agents (e.g., apixaban and rivaroxaban), and lower bleeding profile; however, they are not indicated in some circumstances. For example, DOACs are contraindicated in renal insufficiency (creatinine clearance less than 30 mL/min), antiphospholipid syndrome, and pregnancy, and should be used with caution in liver failure (Ortel et al., *Blood Adv* 2020, 4: 4693-738). Although DOACs have been shown to be effective and safe in the treatment of venous thrombosis at the iliac level, their use in extensive DVTs has not been fully evaluated (Becattini and Agnelli, *Blood* 2020, 135: 305-16). DOACs (rivaroxaban and apixaban) are also recommended for initial treatment of cancer-associated venous thromboembolism, and are preferred, in addition to edoxaban over VKA for long-term anticoagulation due to their improved efficacy. However, caution should be taken when using DOACs in patients with gastrointestinal and potentially genitourinary malignancies due to the increased risk for major bleeding (Key et al., *J Clin Oncol* 2023: Jco2300294). Other anticoagulation agents for the treatment of extremity DVT include low molecular weight heparin (LMWH), fondaparinux, unfractionated heparin (UFH), and VKA. LMWH is the preferred treatment of unprovoked DVTs in pregnancy since DOACs pass through the placenta, and VKA are associated with malformations (Becattini and Agnelli, *Blood* 2020, 135: 305-16). UFH is the preferred treatment if there is a potential need for immediate reversal (e.g., acute bleeding, planned thrombolysis) or if there are contraindications to other agents (e.g., renal insufficiency) (Ortel et al., *Blood Adv* 2020, 4: 4693-738). Medication choice is based on a variety of factors, including comorbidities, patient preference (and affordability), convenience, and whether laboratory monitoring to check for compliance is necessary (Becattini and Agnelli, *Blood* 2020, 135: 305-16; Lenchus et al., *J Intensive Care Med* 2017, 32: 299-311).

Anticoagulation treatment in acute splanchnic vein thrombosis aims to achieve vessel patency and prevent complications such as bowel ischemia or chronic portal hypertension (Valeriani et al., *Vasc Health Risk Manag* 2019, 15: 449-61). In patients with cirrhosis who already have portal hypertension, the goal of treatment is to prevent thrombus progression which may hinder a future liver transplant (Northup et al., *Hepatology* 2021, 73: 366-413). Historically UFH, LMWH, and VKAs have been the anticoagulant agents utilized in the treatment of splanchnic vein thrombosis, however, literature on the use of DOACs is emerging and their usage is becoming more commonplace, specifically in the treatment of PVT. Despite the lack of large clinical trials on the safety and effectiveness of DOACs for the treatment of PVT, the American Association for the Study of Liver Disease includes DOACs as a suggested treatment for PVT but advises caution in patients with advanced portal hypertension (Northup et al., *Hepatology* 2021, 73: 366-413).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the Cochrane Handbook. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources (den Exter et al., *J Thromb Haemost* 2022, 20: 1910-9; Kakkos et al., *Eur J Vasc Endovasc Surg* 2021, 61: 9-82; Stevens et al., *Chest* 2021, 160: e545-e608; Becattini and Agnelli, *Blood* 2020, 135: 305-16; Konstantinides et al., *Eur Heart J* 2020, 41: 543-603; Ortel et al., *Blood Adv* 2020, 4: 4693-738; Mazzolai et al., *Eur Heart J* 2018, 39: 4208-18).

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ^(1, 2, 3, 4)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, One:** ⁽⁵⁾● DVT and anticoagulation (includes PO) and, **Both:** ⁽⁶⁾– Patient or caregiver unable to administer ⁽⁷⁾– Outpatient services unavailable ⁽⁸⁾● **ACUTE, One:** ⁽⁹⁾● Anticoagulation (includes PO) and, **≥ One:** ⁽¹⁰⁾– Extensive DVT and neurovascular assessment at least 6x/24h ^(11, 12)– Splanchnic vein thrombosis ^(13, 14)● UFH, continuous and, **≥ One:** ^(15, 16)– Antiphospholipid antibody syndrome ⁽¹⁷⁾– Bleeding, active or increased risk ⁽¹⁸⁾– BMI ≥ 40 kg/m² or weight > 120 kg ⁽¹⁹⁾– Creatinine clearance ≤ 30 mL/min ⁽²⁰⁾– HIT, suspected or by history and anticoagulation ⁽²¹⁾– IVC filter placement performed ≤ 24 h ⁽²²⁾● **CRITICAL, $\geq$ One:** ⁽²³⁾– Catheter directed thrombolysis scheduled or performed ≤ 24 h ⁽²⁴⁾– Systemic thrombolysis performed ≤ 24 h ⁽²⁴⁾– Hemodynamic monitoring, invasive ⁽²⁵⁾

– Mechanical ventilation

– NIPPV ⁽²⁶⁾– Oxygen $\geq 40\%$ (0.40), ≤ 2 d ⁽²⁷⁾

– Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

● **Episode Day 1, One:** ^(1, 3, 4)

(Symptom or finding within 24h)

● **OBSERVATION, One:** ⁽⁵⁾● DVT and anticoagulation (includes PO) and, **Both:** ⁽⁶⁾– Patient or caregiver unable to administer ⁽⁷⁾– Outpatient services unavailable ⁽⁸⁾● **ACUTE, One:** ⁽⁹⁾● Anticoagulation (includes PO) and, **≥ One:** ^(10, 28)– Extensive DVT and neurovascular assessment at least 6x/24h ^(11, 12)– Splanchnic vein thrombosis ^(13, 14)● UFH, continuous and, **≥ One:** ^(15, 16)– Antiphospholipid antibody syndrome ⁽¹⁷⁾– Bleeding, active or increased risk ⁽¹⁸⁾– BMI ≥ 40 kg/m² or weight > 120 kg ⁽¹⁹⁾– Creatinine clearance ≤ 30 mL/min ⁽²⁰⁾– HIT, suspected or by history and anticoagulation ⁽²¹⁾– Hospital acquired and initiation of anticoagulation (includes PO) ≤ 24 h ^(10, 29)– IVC filter placement performed ≤ 24 h ⁽²²⁾● **CRITICAL, $\geq$ One:** ⁽²³⁾– Catheter directed thrombolysis scheduled or performed ≤ 24 h ⁽²⁴⁾– Systemic thrombolysis performed ≤ 24 h ⁽²⁴⁾– Hemodynamic monitoring, invasive ⁽²⁵⁾

– Mechanical ventilation

– NIPPV ⁽²⁶⁾– Oxygen $\geq 40\%$ (0.40), ≤ 2 d ⁽²⁷⁾

– Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

● **Episode Day 2, One:** (3, 4, 30)

● **OBSERVATION, One:**

● **Responder**, discharge expected today if clinically stable last 12h, **Both:** (31)

- Anticoagulation regimen established (32, 33)
- Coordination of outpatient services completed (34)

● **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**

- Anticoagulation (includes PO) and, **Both:** (6)
 - Patient or caregiver unable to administer (7)
 - Outpatient services unavailable (8)

● **ACUTE, One:**

● **Early responder**, discharge expected today if clinically stable last 12h, **Both:** (31)

● Anticoagulation regimen, **One:**

- Established (32)
- Contraindicated (35)
- No complication or active comorbidity relevant to this episode of care (36)

● **Partial responder**, not clinically stable for discharge and requires continued stay, **≥ One:**

– Embolectomy or thrombectomy (**use General Surgical subset**)

● Anticoagulation (includes PO) and, **≥ One:** (10, 28)

- Extensive DVT and neurovascular assessment at least 6x/24h (11, 12)
- Splanchnic vein thrombosis (13, 14)

● UFH, continuous and, **≥ One:** (15, 16)

- Antiphospholipid antibody syndrome (17)
- Bleeding, active or increased risk (18)
- BMI ≥ 40 kg/m² or weight > 120 kg (19)
- Creatinine clearance ≤ 30 mL/min (20)
- HIT, suspected or by history and anticoagulation (21)
- IVC filter placement performed ≤ 24 h (22)

● Bleeding, actual or suspected and, **≥ One:** (37, 38)

- Blood product transfusion
- Hct or Hb monitoring at least 2x/24h, ≤ 2 d
- Post Critical care ≤ 24 h

● **CRITICAL, $\geq$ One:**

– Catheter directed thrombolysis scheduled or performed ≤ 24 h (24)

– Systemic thrombolysis performed ≤ 24 h (24)

– Post thrombolytic therapy ≤ 24 h

– Hemodynamic monitoring, invasive (25)

– HFNC and respiratory monitoring every 1-2h (27, 39, 40)

– Mechanical ventilation

● NIPPV and, **≥ One:** (26)

- FiO₂ > 50%(0.50)
- Respiratory intervention every 1-2h (41)

– Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

● **Episode Day 3, One:** (4, 42, 43)

● **OBSERVATION, One:**

● **Responder**, discharge expected today if clinically stable last 12h, **Both:** (31)

- Anticoagulation regimen established (32, 33)
- Coordination of outpatient services completed (34)

● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**

- For DVT use Episode Day 3 at a higher level of care
- For a condition other than DVT use appropriate subset (Episode Day 1) based on clinical findings
- Refer for secondary review

● **ACUTE, One:**● **Responder**, discharge expected today if clinically stable last 12h, **Both:** ⁽³¹⁾● **Anticoagulation regimen, One:**

- Established ⁽³²⁾
- Contraindicated ⁽³⁵⁾

● **Complication or comorbidity, One:**

- No complication or active comorbidity relevant to this episode of care ⁽³⁶⁾

● **Complication or comorbidity and clinically stable for discharge, ≥ One:**

- Bleeding controlled or resolved
- Laboratory values within acceptable limits

● **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**

- Embolectomy or thrombectomy (**use General Surgical subset**)

● **Anticoagulation (includes PO) and, ≥ One:** ^(10, 28)

- Extensive DVT and neurovascular assessment at least 6x/24h ^(11, 12)
- Splanchnic vein thrombosis ^(13, 14)

● **UFH, continuous and, ≥ One:** ^(15, 16)

- Antiphospholipid antibody syndrome ⁽¹⁷⁾
- Bleeding, active or increased risk ⁽¹⁸⁾
- BMI ≥ 40 kg/m² or weight > 120 kg ⁽¹⁹⁾
- Creatinine clearance ≤ 30 mL/min ⁽²⁰⁾

- HIT and anticoagulation ^(21, 44)

- IVC filter placement performed ≤ 24h ⁽²²⁾

● **Pretreatment or bridging anticoagulation, All:** ⁽⁴⁵⁾

- Patient or caregiver unable to administer
- Outpatient services unavailable ⁽⁸⁾

● **Intervention, One:**

- INR non-therapeutic and anticoagulation (includes PO) ⁽⁴⁶⁾
- Anticoagulation ≤ 5d since initiation ^(47, 48)

● **Bleeding, actual or suspected and, ≥ One:** ^(37, 38)

- Blood product transfusion
- Hct or Hb monitoring at least 2x/24h, ≤ 2d
- Post Critical care ≤ 24h

● **CRITICAL, ≥ One:**

- Post thrombolytic therapy ≤ 24h ⁽²⁴⁾
- Hemodynamic monitoring, invasive ⁽²⁵⁾
- HFNC and respiratory monitoring every 1-2h ^(27, 39, 40)
- Mechanical ventilation

● **NIPPV and, ≥ One:** ⁽²⁶⁾

- FiO₂ > 50%(0.50)
- Respiratory intervention every 1-2h ⁽⁴¹⁾

- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

● **Episode Day 4, One:**● **ACUTE, One:**● **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³¹⁾● **Anticoagulation regimen, One:**

- Established ⁽³²⁾
- Contraindicated ⁽³⁵⁾

- Coordination of outpatient services completed or not indicated ⁽³⁴⁾

● **Complication or comorbidity, One:**

- No complication or active comorbidity relevant to this episode of care ⁽³⁶⁾

● **Complication or comorbidity and clinically stable for discharge, ≥ One:**

- Bleeding controlled or resolved
- Laboratory values within acceptable limits

● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One**:

- Use **Extended Stay subset**
- Refer for secondary review

● **CRITICAL, One**:

● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One**:

- Use **Extended Stay subset**
- Refer for secondary review

Discharge, Level of Care, One: ⁽⁴⁹⁾

● **HOME, Both**:

● Level of care appropriateness, **Both**:

- Home environment safe and accessible ⁽⁵⁰⁾
- Patient or caregiver demonstrates ability to manage care

● DVT discharge planning, **All**:

- Provide comprehensive written discharge and teaching instructions ⁽⁵¹⁾

● Education on the importance of, **All**:

- Anticoagulation management ⁽⁵²⁾
- Caution when using nonsteroidal anti-inflammatory medications or aspirin ⁽⁵³⁾
- DVT prevention methods ⁽⁵⁴⁾
- Medication reconciliation ⁽⁵⁵⁾
- Follow-up care planned ⁽⁵⁶⁾
- Identify and address transportation needs

● **HOME CARE, Both**:

● Level of care appropriateness, **All**:

- Home environment safe and accessible ⁽⁵⁰⁾
- Patient or caregiver willing and able to learn care
- Outpatient management contraindicated or unavailable ^(57, 58)

● Skilled services, **≥ One**: ⁽⁵⁹⁾

- Skilled nursing ⁽⁶⁰⁾
- Skilled therapy, PT, OT, or ST
- Behavioral health ⁽⁶¹⁾

● DVT discharge planning, **All**:

- Provide comprehensive written discharge and teaching instructions ⁽⁵¹⁾

● Education on the importance of, **Both**:

- Anticoagulation management ⁽⁵²⁾
- DVT prevention methods ⁽⁵⁴⁾
- Medication reconciliation ⁽⁵⁵⁾
- Follow-up care planned ⁽⁵⁶⁾
- Identify and address transportation needs

● **BEHAVIORAL HEALTH, Both**:

● Level of care appropriateness, **One**:

- Support systems available ⁽⁶²⁾

● Skilled treatment, **≥ One**:

- Clinical assessment ⁽⁶³⁾
- Individual, group, or family counseling
- Psychiatric, substance, or medication evaluation ⁽⁶⁴⁾

● **SKILLED MEDICAL OR THERAPY, Both**:

● Level of care appropriateness, **All**:

- Medical practitioner, NP, PA assessment or oversight $\geq 1x/wk$
- Treatment precluded in a lower level

● Services, **≥ One**:

- Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁶⁰⁾
- **Therapy, All:**
 - Goal directed therapy with rehabilitation potential ^(65, 66)
 - Functional impairments requiring at least supervision ⁽⁶⁷⁾
 - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
- **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁵⁵⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
 - **Level of care appropriateness, All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
 - Treatment precluded in a lower level
 - **Services, ≥ One:**
 - Medical and skilled nursing interventions or assessment 4h/24h ⁽⁶⁰⁾
 - **Therapy, All:**
 - Goal directed therapy with rehabilitation potential ^(65, 66)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁶⁷⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁵⁵⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE REHAB, Both:**
 - **Level of care appropriateness, All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - Treatment precluded in a lower level
 - **Services, All:**
 - Goal directed therapy with rehabilitation potential ^(65, 66)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁶⁷⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁵⁵⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **ACUTE REHAB, Both:**
 - **Level of care appropriateness, Both:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - **Services, All:**
 - Goal directed therapy with rehabilitation potential ^(65, 66)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁶⁷⁾
 - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines ⁽⁶⁸⁾
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁵⁵⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
 - **Level of care appropriateness, Both:**
 - Treatment precluded in a lower level

- In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One**:
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽⁶⁹⁾
 - Ventilator dependent ≥ 6h/d and weaning planned ^(70, 71)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽⁷²⁾
- Complete prior to facility transfer, **All**:
 - Medication reconciliation ⁽⁵⁵⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
294 DEEP VEIN THROMBOPHLEBITIS WITH CC/MCC	n/a	3.5	CMS GMLOS
295 DEEP VEIN THROMBOPHLEBITIS WITHOUT CC/MCC	n/a	2.4	CMS GMLOS
299 PERIPHERAL VASCULAR DISORDERS WITH MCC	n/a	4	CMS GMLOS
300 PERIPHERAL VASCULAR DISORDERS WITH CC	n/a	3.1	CMS GMLOS
301 PERIPHERAL VASCULAR DISORDERS WITHOUT CC/MCC	n/a	2.1	CMS GMLOS
393 OTHER DIGESTIVE SYSTEM DIAGNOSES WITH MCC	n/a	4.4	CMS GMLOS
394 OTHER DIGESTIVE SYSTEM DIAGNOSES WITH CC	n/a	3	CMS GMLOS
395 OTHER DIGESTIVE SYSTEM DIAGNOSES WITHOUT CC/MCC	n/a	2.1	CMS GMLOS
441 DISORDERS OF LIVER EXCEPT MALIGNANCY, CIRRHOSIS OR ALCOHOLIC HEPATITIS WITH MCC	n/a	4.8	CMS GMLOS
442 DISORDERS OF LIVER EXCEPT MALIGNANCY, CIRRHOSIS OR ALCOHOLIC HEPATITIS WITH CC	n/a	3.2	CMS GMLOS
443 DISORDERS OF LIVER EXCEPT MALIGNANCY, CIRRHOSIS OR ALCOHOLIC HEPATITIS WITHOUT CC/MCC	n/a	2.5	CMS GMLOS
DEEP VEIN THROMBOSIS	29%	3.6	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

3:

Expected progress after 12-24h

- Anticoagulation therapy initiated

Consider:

- Anticoagulation medication regimen education
- Discharge planning

4:

Care Facilitation

- Prolonged anticoagulation therapy expected, arrange home monitoring if indicated (**Home, Home Care**)
- Medication adherence to anticoagulant regimen (**Home Care**)
- Potential risk for readmission (Minges et al., Am J Med 2018, 131: 1200-8) (**Home Care, Readmission Reduction Program**)
- Knowledge deficit or non-adherence with condition management identified (**Home Care, Readmission Reduction Program**). Provide patient education including anticoagulation medication regimen education, nutritional teaching for patients prescribed warfarin, and over-the-counter medications to avoid (e.g., nonsteroidal anti-inflammatory drugs, aspirin).
- Subtherapeutic anticoagulation and arrange anticoagulation monitoring (**Home Care**)

Social Determinants of Health (SDOH)

Social concerns may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. The following social determinants of health should be evaluated on admission to determine which factors are relevant to an individual patient and condition. This list may not be all inclusive, and all relevant SDOH factors should be considered. Any SDOH factors determined to be relevant to a given patient should be planned for throughout the hospital stay.

Alcohol or substance use

- Care coordination with outpatient medical or behavioral health providers
- Information should be given about community resources and programs
- Referral to addiction social support systems

Domestic violence and sexually abused individuals

- Community resources and legal information should be given
- Create a long-term safety plan or make appropriate referral

End of life planning, arrange palliative care or hospice

- Consider palliative care for patients with end-stage disease
- Consider hospice care for terminally ill patients with a life expectancy of less than 6 months

Financial resources strained or limited

- For patients with income insecurity, identify income-support resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Consider direct income assistance benefits and indirect income assistance programs (i.e., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)
- Consider referral to Area Agencies on Aging (AAA) for information on nutrition and meals programs, homemaking services, transportation resources
- For food insecurity, provide information about local food banks and Supplemental Nutrition Assistance Program (SNAP) incentive programs (Berman et al., Pediatr Rev 2018, 39: 235-46)

Functional or activities of daily living (ADL) impairment

- Consider an alternate level of care
- If patient is unable to make decision for self, refuses treatment, or demands to be discharged without a rational reason, a capacity evaluation may be needed to ensure patient's ability to make decisions for self
- Information should be given to patient and family about Advance Directive or Power of Attorney

Mental health co-morbidities involved

- Appointment with primary care provider or behavioral health specialist upon discharge
- Care coordination with outpatient behavioral health providers
- Community resources should be given

Risk for readmission

- Consider home care services
- Consider transfer to an alternative level of care or readmission reduction program
- Provide patient and family with information on disease management programs
- Consider telehealth and mobile health interventions

Risk for treatment nonadherence

- For patients with poor insight into illness or noncompliance with treatment, introduce peer support groups to help improve service user experience and quality of life (National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.)
- In-home intervention to prevent medical relapse, or readmission, and to enhance medication compliance
- For patients at risk of medication nonadherence, consider patient education and behavioral support initiatives such as text message reminders to take a medication and electronic medication dispensers

Environmental exposure to toxic substances and other physical hazards

- For tobacco use and/or exposure, information should be given about tobacco-use cessation and the risk of second-hand smoke
- For lead poisoning, provide community resources for primary prevention and secondary prevention strategies (i.e., blood lead level testing and follow up) (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)

Unstable living environment or homelessness expected upon discharge

- Health insurance application and information should be given to patient
- Homeless shelter application and information should be given to patient
- Referral to walk-in state or federally sponsored outpatient programs
- Ensure access to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
- For individuals with homelessness and the greatest service need (i.e., serious mental illness, history of hospitalizations, functional impairment), consider Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community-Based Treatment (Ponka et al., PLoS One 2020, 15: e0230896)

Unsafe home environment

- If home environment is unsafe, consider a skilled nursing facility (SNF)
- For patients at risk for falls, provide information on community-based fall prevention programs through the National Council on Aging

5:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

6:

Anticoagulation may include the following:

- Direct oral anticoagulants (DOACs)
- Low molecular weight heparin (LMWH)
- Fondaparinux

7:

Although the American Society of Hematology (ASH) guidelines for management of venous thromboembolism recommend offering home treatment over hospital treatment for uncomplicated deep vein thrombosis (DVT), this recommendation does not apply to individuals who have inadequate home support, lack of access to medication, or have a history of poor compliance (Ortel et al., Blood Adv 2020, 4: 4693-738). An individual's social circumstances may hinder outpatient DVT management; therefore, these circumstances should be considered when the discharge disposition of a patient is being assessed (Kakkos et al., Eur J Vasc Endovasc Surg 2021, 61: 9-82).

8:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting
- Skilled nursing services unavailable (e.g., Home Care)

9:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

10:

Anticoagulation may include the following:

- Direct oral anticoagulants (DOACs)
- Low molecular weight heparin (LMWH)
- Unfractionated heparin (UFH), continuous
- Fondaparinux

11:

Extensive deep vein thrombosis (DVT) includes a thrombus involving the iliofemoral vein, multiple veins, and/or causes severe symptoms (e.g., swelling of the entire limb, phlegmasia dolens). The incidence of limb-threatening disease has been shown to be higher in iliofemoral DVTs compared to popliteal or isolated distal DVTs (Becattini and Agnelli, Blood 2020, 135: 305-16). Patients with extensive DVTs may require hospitalization at the Acute level of care for neurovascular assessments due to the risk for limb ischemia and potential need for thrombolytic therapy. Unfractionated heparin (UFH) is indicated for patients being considered for thrombolysis (Ortel et al., Blood Adv 2020, 4: 4693-738).

12:

A neurovascular assessment is done to identify any signs of neurovascular compromise and should include the following:

- Evaluation of the quality of the peripheral pulses and capillary refill rate
- Pain assessment
- Assessment for warmth in the extremity
- Gross motor and sensory function of fingertips and toes

Presence of any of the following signs suggests neurovascular compromise: painful passive motion, pallor of the extremity, paresthesias, paralysis, pulselessness or diminished pulses, or poikilothermia (fluctuating temperature) (Gerhard-Herman et al., Circulation 2017, 135: e686-e725).

13:

Splanchnic vein thrombosis refers to a thrombus in one or more of the following locations:

- Portal vein
- Mesenteric vein
- Splenic vein

14:

The 2018 American Association for the Study of Liver Disease (AASLD) guidelines suggest DOACs as a treatment option for portal vein thrombosis (PVT) based on small cohort studies and advise caution in their use in patients with cirrhosis (Northup et al., Hepatology 2021, 73: 366-413). Since then, more studies have been conducted showing that the use of DOACs in liver disease may be as safe and effective as other anticoagulants. A recent systematic review and meta-analysis (including 3 studies comparing DOACs to warfarin) found that the DOAC group had lower thrombus extension rates, higher recanalization rates, and no difference in bleeding rates compared to the warfarin group in patients with cirrhosis who have PVT (Zhang et al., Clin Appl Thromb Hemost 2022, 28: 10760296221104797). Despite the emerging literature showing DOACs may be safe in patients with liver disease, more research on the safety and efficacy of these agents is needed. In addition, some DOAC agents (e.g., rivaroxaban) are contraindicated in moderate liver impairment (e.g., Child-Pugh class B), and all DOACs should not be used in patients with severe liver impairment (e.g., Child-Pugh class C) (Chun et al., Clin Mol Hepatol 2021, 27: 535-52).

15:

Instruction: This line of criteria includes continuous infusion and rate adjustments to achieve a targeted therapeutic goal and excludes prophylactic dosages.

16:

Unfractionated heparin (UFH) is preferable over low molecular weight heparin (LMWH) for the treatment of venous thromboembolism (VTE) in the following conditions (Mazzolai et al., Eur Heart J 2018, 39: 4208-18):

- Antiphospholipid antibody syndrome
- Bleeding or risk for bleeding
- BMI ≥ 40 kg/m² or weight > 120 kg
- Renal failure (CrCl < 30 mL/min)

17:

Thrombotic events are common in patients with antiphospholipid antibody syndrome. The ease of use and lack of need for monitoring have led to interest in using direct oral anticoagulants (DOACs) for management, but warfarin continues to be preferred by many providers. While the use of low molecular weight heparin (LMWH) in bridging to warfarin may be appropriate for some patients, unfractionated heparin is preferred in patients with known or suspected catastrophic antiphospholipid syndrome, or in patients with anticoagulant-refractory disease who may require higher intensity dosing (and as such, would incur an increased risk of bleeding) (Cohen et al., J Thromb Haemost 2021, 19: 892-908).

18:

The risk for major bleeding is highest during the first month of anticoagulant therapy, making it imperative that a patient's risk for bleeding be assessed at the time of initiation of anticoagulant treatment (Konstantinides et al., Eur Heart J 2020, 41: 543-603). In patients with an increased risk for bleeding, unfractionated heparin (UFH) may be the preferred initial treatment due to its short half-life and reversibility with intravenous (IV) protamine sulfate (Mazzolai et al., Eur Heart J 2018, 39: 4208-18). Although there are several validated venous thromboembolism (VTE) bleeding risk assessment tools, studies have yet to be conducted to determine their impact on treatment decisions; therefore, the International Society of Thrombosis and Haemostasis (ISTH) did not make a recommendation on which bleeding risk score to use prior to starting anticoagulation therapy (den Exter et al., J Thromb Haemost 2022, 20: 1910-9). As a result, the bleeding risk factors for this criterion were derived from a combination of the best-studied risk assessment tools for major bleeding in patients with VTE, guidelines, and literature focusing on a moderate to strong association with the risk of bleeding (den Exter et al., J Thromb Haemost 2022, 20: 1910-9; Klok and Huisman, Blood 2020, 135: 724-34; Konstantinides et al., Eur Heart J 2020, 41: 543-603; Brown et al., J Am Heart Assoc 2018, 7: epub; Kearon et al., Chest 2016, 149: 315-52).

Instruction:

A patient who has a scheduled or anticipated procedure or surgery and/or has one or more of the following risk factors meets for this criterion:

- Major bleeding within the last 3 months (e.g., bleeding that caused a fall in hemoglobin ≥ 2 g/dL, blood transfusion of ≥ 2 units, or symptomatic bleeding in a critical area or organ (e.g., intracranial, spinal, ocular, pericardia, articular, intramuscular with compartment syndrome, retroperitoneal)) (den Exter et al., J Thromb Haemost 2022, 20: 1910-9; Konstantinides et al., Eur Heart J 2020, 41: 543-603; Brown et al., J Am Heart Assoc 2018, 7: epub; Klok et al., Thromb Haemost 2017, 117: 1164-70; Kearon et al., Chest 2016, 149: 315-52)
- Renal failure (e.g., creatinine clearance 30-60 mL/min, receiving dialysis) (den Exter et al., J Thromb Haemost 2022, 20: 1910-9; Konstantinides et al., Eur Heart J 2020, 41: 543-603; Brown et al., J Am Heart Assoc 2018, 7: epub; Klok et al., Thromb Haemost 2017, 117: 1164-70; Kearon et al., Chest 2016, 149: 315-52; Ruiz-Giménez et al., Thromb Haemost 2008, 100: 26-31)
- Active cancer (e.g., receiving treatment) (den Exter et al., J Thromb Haemost 2022, 20: 1910-9; Klok et al., Thromb Haemost 2017, 117: 1164-70; Konstantinides et al., J Am Coll Cardiol 2016, 67: 976-90; Kearon et al., Chest 2008, 133: 454S-545S; Ruiz-Giménez et al., Thromb Haemost 2008, 100: 26-31)
- Anemia (e.g., hemoglobin < 13 g/dL in male sex assigned at birth, < 12 g/dL in female sex assigned at birth) (den Exter et al., J Thromb Haemost 2022, 20: 1910-9; Klok et al., Thromb Haemost 2017, 117: 1164-70; Kearon et al., Chest 2016, 149: 315-52; Ruiz-Giménez et al., Thromb Haemost 2008, 100: 26-31)
- Age > 75 years (Konstantinides et al., Eur Heart J 2020, 41: 543-603)
- Concurrent use of platelet inhibitors or nonsteroidal anti-inflammatory drugs (NSAIDs) (den Exter et al., J Thromb Haemost 2022, 20: 1910-9; Konstantinides et al., Eur Heart J 2020, 41: 543-603; Brown et al., J Am Heart Assoc 2018, 7: epub; Kearon et al., Chest 2016, 149: 315-52)
- Uncontrolled hypertension (e.g., systolic blood pressure ≥ 140 mmHg) (den Exter et al., J Thromb Haemost 2022, 20: 1910-9; Brown et al., J Am Heart Assoc 2018, 7: epub; Klok et al., Thromb Haemost 2017, 117: 1164-70)
- Thrombocytopenia (e.g., platelets $< 100,000$ cu.mm(100×10^9 /L)) (den Exter et al., J Thromb Haemost 2022, 20: 1910-9; Kearon et al., Chest 2016, 149: 315-52)
- Liver failure (Konstantinides et al., Eur Heart J 2020, 41: 543-603; Brown et al., J Am Heart Assoc 2018, 7: epub; Kearon et al., Chest 2016, 149: 315-52)
- Previous stroke (hemorrhagic or ischemic), timeframe not defined by literature (Konstantinides et al., Eur Heart J 2020, 41: 543-603; Brown et al., J Am Heart Assoc 2018, 7: epub; Kearon et al., Chest 2016, 149: 315-52)
- Recent surgery (e.g., within the last 3 months) (Kearon et al., Chest 2016, 149: 315-52)

19:

Unfractionated heparin is often used in the management of acute thromboembolic events in patients with a body mass index (BMI) greater than or equal to 40 kg/m² or weight greater than 120 kg; this practice is based on mixed evidence for direct oral anticoagulants (DOACs) in this population, and concerns about low molecular weight heparin (LMWH) bioavailability. The 2016 International Society on Thrombosis and Haemostasis (ISTH) guidelines recommended against the use of DOACs in patients with a BMI greater than or equal to 40 kg/m² or weight greater than 120 kg, and a 2019 systematic review (19 studies, including pharmacokinetic, studies, phase III RCTs and post hoc analyses, and retrospective cohort studies) continued to suggest caution (Kido et al., Pharmacotherapy 2020, 40: 72-83); however, a recent meta-analysis (5 observational studies, n=6585) concluded that these agents were non-inferior to warfarin (Elshafei et al., J Thromb Thrombolysis 2021, 51: 388-96). Absorption of LMWH administered subcutaneously may be variable in patients with obesity, and treatment with intravenous unfractionated heparin provides a reliable means of anticoagulation for patients with acute thromboembolic events (Lee et al., Clin Drug Investig 2020, 40: 33-40).

20:

Patients with creatinine clearance less than or equal to 30 mL/min should be treated with unfractionated heparin (UFH) when parenteral anticoagulation is necessary. Low molecular weight heparin (LMWH) and certain direct oral anticoagulants are predominantly excreted by the kidneys, and therefore may accumulate in patients with renal insufficiency causing an increased risk of bleeding (Kakkos et al., Eur J Vasc Endovasc Surg 2021, 61: 9-82).

21:

Heparin-induced thrombocytopenia (HIT) results from sensitization to the heparin molecule. Usually, it occurs between 5 to 10 days after the current heparin exposure but may occur as early as minutes to hours after exposure. HIT is a greater than 50%(0.50) serial reduction in a patient's baseline platelet count. A direct thrombin inhibitor (DTI) (e.g., argatroban, bivalirudin) is the initial treatment for patients with active HIT, a history of HIT, or those

with a strong clinical suspicion of HIT who require anticoagulation. Although the American Society of Hematology (ASH) guidelines suggest the use of fondaparinux or direct oral anticoagulants (DOACs), the recommendation is conditional and based on very low certainty of evidence. In addition, these agents are considered off-label use in the treatment of HIT since they have yet to be approved by the U.S. Food and Drug Administration (FDA) for this condition. For patients who require long-term anticoagulation, warfarin should not be initiated until platelet count recovers (e.g., platelet count greater than $150 \times 10^9/L$). Additionally, a 5-day overlap with warfarin and DTI therapy is recommended (Cuker et al., Blood Adv 2018, 2: 3360-92; Guyatt et al., Chest 2012, 141: 48S-52S).

22:

When anticoagulation treatment for a deep vein thrombosis (DVT) is contraindicated and there is risk for pulmonary embolism (PE), an inferior vena cava (IVC) filter may be considered. Although useful in prevention of PE, there are associated risks with IVC filter placement. These risks include bleeding during placement, embolization or perforation of the filter, and risk of recurrent DVT (Stevens et al., Chest 2021, 160: e545-e608; Kaufman et al., J Vasc Interv Radiol 2020, 31: 1529-44).

23:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

24:

Thrombolytic therapy is reserved for limb-threatening deep vein thrombosis (DVT) and may also be considered for treatment of symptomatic iliac and common femoral DVTs in young patients with minimal risk for bleeding to prevent severe post-thrombotic syndrome. Although evidence is lacking comparing catheter-directed versus systemic thrombolysis for treatment of extensive DVT, guidelines recommend using catheter-directed over systemic thrombolysis due to the known bleeding risks associated with systemic thrombolysis (Ortel et al., Blood Adv 2020, 4: 4693-738). Given the low quality of evidence guiding this recommendation, systemic thrombolysis remains an option in criteria.

Thrombolytic therapy should also be considered in patients with mesenteric vein thrombosis that do not respond to anticoagulation therapy and are at risk for intestinal ischemia. In a small observational study, the use of streptokinase infusion (mostly administered via the transhepatic route) showed successful recanalization of the mesenteric vein in 19 out of 20 patients (Simonetto et al., Am J Gastroenterol 2020, 115: 18-40).

25:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

26:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in

a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

27:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS**Nasal cannula**

Room air 21%(0.21)

1L 24%(0.24) **4L** 36%(0.36)

2L 28%(0.28) **5L** 40%(0.40)

3L 32%(0.32) **6L** 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS**Air-entrainment masks/nebulizers**

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

28:

Although warfarin requires bridging with a parenteral anticoagulant while the vitamin K-dependent clotting factors are being depleted, it should be initiated on day 1 or 2 of parenteral anticoagulation therapy in patients with thromboembolic disease, unless contraindicated (Trang et al., AACN Adv Crit Care 2019, 30: 209-16; Holbrook et al., Chest 2012, 141: e152S-84S).

29:

Hospital acquired refers to a condition that developed during the patient's current hospitalization. Criteria for hospital acquired conditions cannot be applied to patients who either have been discharged to home or transferred from a post-acute facility.

30:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

31:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

32:

An outpatient anticoagulation regimen may include low molecular weight heparin (LMWH), fondaparinux, warfarin, or a direct oral anticoagulant (DOAC). For patients who will be treated with warfarin, initiation of treatment must overlap with parenteral anticoagulation until a therapeutic international normalized ratio (INR) is achieved; therefore, some patients could be discharged on parenteral anticoagulant along with warfarin if discharged prior to completion of overlap therapy. Dabigatran and edoxaban, both DOACs, require 5 to 10 days of pretreatment with parenteral anticoagulation; therefore, these patients may be discharged on a parenteral anticoagulation with the plan to switch to dabigatran or edoxaban after pretreatment is complete (Ortel et al., Blood Adv 2020, 4: 4693-738). This criterion also includes patients with confirmed or a history of heparin-induced thrombocytopenia (HIT) who have achieved therapeutic anticoagulation and whose outpatient regimen has been established.

33:

Home therapy with low molecular weight heparin (LMWH) has been shown to be as safe as inpatient administration of LMWH or unfractionated heparin for patients with uncomplicated deep vein thrombosis (DVT) (Ortel et al., Blood Adv 2020, 4: 4693-738).

34:

The American Society of Hematology (ASH) recommends that patients treated with anticoagulation therapy for a venous thromboembolism receive specialized anticoagulation management services rather than care provided by their regular health care provider (Witt et al., Blood Adv 2018, 2: 3257-91).

35:

Although anticoagulation is the recommended treatment for venous thromboembolism (VTE) (e.g., deep vein thrombosis (DVT), pulmonary embolism (PE)), there are some circumstances in which anticoagulation may be contraindicated (e.g., active bleeding). If a patient cannot be treated with anticoagulation, a retrievable inferior vena cava (IVC) filter may be inserted, and the patient could be discharged without anticoagulation (Ortel et al., Blood Adv 2020, 4: 4693-738). Patients who experience an episode of major bleeding while on anticoagulation therapy may also be discharged without anticoagulation treatment initially, since guidelines recommend resumption of oral anticoagulation within 90 days (Witt et al., Blood Adv 2018, 2: 3257-91).

36:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

37:

Patients who develop bleeding while receiving anticoagulation should be evaluated for an underlying source of bleeding. Evaluation may include upper and lower gastrointestinal (GI) studies to rule out underlying GI pathology or an imaging study to rule out intracranial bleeding (Janz, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 9 edn. 2018:1463-79.e3).

38:

Findings associated with bleeding may include (Janz, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 9 edn. 2018:1463-79.e3):

- Heart rate 100-120/min, sustained
- Hct or Hb decreased within last 24h

- Orthostatic hypotension
- Mental status change
- Presyncope or syncope

39:

High-flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO_2) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the FiO_2 . The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high-flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (i.e., face mask or nasal cannula with a flow rate of 15 L/min); however, it did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has been shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

40:

Respiratory monitoring includes an evaluation of the patient's respiratory rate, heart rate, oxygen saturation, or work of breathing. Within 90 minutes of high-flow nasal cannula (HFNC) initiation, the patient's respiratory status should improve, indicating adequate respiratory support is being provided. If clinical improvement is not achieved within this time frame, additional respiratory support may be indicated (Lee et al., Intensive Care Med 2013, 39: 247-57).

41:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

42:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status**; those who do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

43:**Expected progress after 48-72h**

- Anticoagulation progressing, if on warfarin, international normalized ratio (INR) increasing towards the therapeutic goal
- No evidence of bleeding
- Preparing for discharge when INR is therapeutic or when anticoagulation teaching is completed

If no, consider:

- Adjusting dosage or changing anticoagulant agent and/or combination therapy, possible hematology or gastroenterology consultation

44:

Instruction: This criterion applies to patients with actual, suspected, or a history of heparin-induced thrombocytopenia (HIT).

45:

Although the American Society of Hematology (ASH) guidelines for management of venous thromboembolism (VTE) recommend offering home treatment over hospital treatment for uncomplicated deep vein thrombosis (DVT), this recommendation does not apply to individuals who have inadequate home support, lack of access to medication, or a history of poor compliance (Ortel et al., Blood Adv 2020, 4: 4693-738). Patients with these risk factors may require inpatient admission for bridging or pre-treatment with parenteral anticoagulation (e.g., fondaparinux, low molecular weight heparin) prior to being converted to an oral regimen.

46:

Instruction: Patients who have not achieved a therapeutic INR on 2 consecutive days and continue on LMWH or fondaparinux will meet this criterion.

According to the American Society of Hematology (ASH), when warfarin is being initiated, it should be overlapped with parenteral anticoagulation (e.g., fondaparinux, low molecular weight heparin (LMWH)) until a therapeutic international normalized ratio (INR) of 2.0-3.0 is achieved (Ortel et al., Blood Adv 2020, 4: 4693-738).

47:

Patients to be treated with dabigatran or edoxaban, both direct oral anticoagulants (DOACs), require at least 5 days of parenteral anticoagulation prior to initiation (Ortel et al., Blood Adv 2020, 4: 4693-738).

Instruction: This criteria point is intended for patients who require continued hospitalization for pretreatment with parenteral anticoagulation (e.g., low molecular weight heparin (LMWH) or fondaparinux) prior to starting dabigatran or edoxaban.

48:

Anticoagulation includes low molecular weight heparin (LMWH) or fondaparinux.

49:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

50:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

51:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

52:

Anticoagulants are among the most common drugs associated with medication errors. Before discharge, patients and/or caregivers should receive anticoagulant counseling tailored to their unique medication and delivered at an appropriate literacy level. The Joint Commission recommends that anticoagulation education cover the importance of follow-up monitoring, adherence, drug-food interactions, and the risk of adverse effects and drug interactions (Hawes, Pharmacy (Basel) 2018, 6: epub).

53:

The use of nonsteroidal anti-inflammatory medications or aspirin while on anticoagulant therapy has been associated with an increased risk for bleeding (den Exter et al., J Thromb Haemost 2022, 20: 1910-9).

54:

Mitigating risk factors (e.g., discontinuing hormone medication, smoking cessation) can aid in the prevention of future venous thromboembolism (Mount et al., Am Fam Physician 2022, 105: 377-85; Chopard et al., Jama 2020, 324: 1765-76).

55:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

56:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

57:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

58:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

59:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

60:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

61:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

62:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

63:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

64:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

65:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

66:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

67:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

68:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

69:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

70:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

71:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6;

Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

72:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.